

1 Topics

This exam covers everything in the course. Here is a list of topics covered so far in the course. Last updated: **July 6, 2026**

1. Ordered Fields
 - Examples/Non-examples
 - Definition and properties of $\leq$
2. The Real Line
 - Definition and properties of the absolute value
 - Definition and properties of sup/inf
 - The completeness axiom
 - Archimedean property
 - Dense sets
3. Metric Spaces
 - Examples of metric spaces.
 - Norms induce metrics
 - Open/Closed sets in metric spaces
 - Bounded sets.
 - Interior, Closure, and Boundaries
 - Equivalent norms and metrics
4. Sequences in Metric Spaces
 - Bounded and convergent subsequences
 - Subsequences
5. Sequences in $\mathbb{R}$
 - Properties of convergent subsequences
 - Monotone sequences
 - Definition and properties of limsup/liminf
6. Completed Metric Spaces
 - Cauchy sequences
 - Definition of a complete metric space
 - $\mathbb{R}$ is complete
 - $\mathbb{R}^d$ is complete
7. Topological Spaces
 - Definition of a Topological Space $(X, \mathcal{T})$.
 - Metric Topology
 - Convergence in Topological spaces
 - Definition compact sets
 - Closed $\cap$ compact is closed
 - Nested sequences of compact sets have non-empty compact intersection.

8. Compact Sets in Metric Spaces.

- Definition of Sequentially compact
- Definition of Totally Bounded
- examples and non-examples of seq
- Compact $\implies$ Seq. Compact $\implies$ Totally Bounded and Complete $\implies$ Compact.
- Heine-Borel Theorem.

9. Continuous functions

- Definition and equivalent definitions
- Cluster points
- Images of compact sets are compact
- Properties of real-values continuous functions (e.g. $f + g$ is continuous.)

10. Uniform continuity.

- Definition
- Examples/Non-examples
- For uniformly continuous f , x_n is Cauchy $\implies f(x_n)$ is Cauchy.
- Continuous functions with compact domain are uniformly continuous.

11. Sequences of functions and types of convergence

- Definition of Point-wise convergence
- Example where the limit of sequence of functions is not continuous.
- Definition of uniform convergence
- Example of uniformly converging sequence of functions.
- Uniform Cauchy Criterion
- For complete co-domains, Uniformly Cauchy $\iff$ Uniform convergence

12. Connected sets

- Definitions
- $[0, 1]$ is connected.
- Definition of path connected.
- Images of connected sets are connected.
- Examples of path connected and connected sets.
- Path connected implies connected. Converse is not true.
- Characterization of connected sets in $\mathbb{R}$.
- Open and connected in $\mathbb{R}^n$ implies path connected.
- The topologist's curve. (Do NOT need to know **how** to show it is connected but not path connected)

13. Differentiation

- Definition of the derivative
- Differentiation rules
- Mean Value Theorem and Rolle's Theorem
- Corollaries of MVT and Rolle's
- Inverse function theorem
- Taylor's theorem and Corollaries

14. Integration

- Partitions
- Upper and lower Darboux sums
- Darboux integrable
- Cauchy Criterion for integrals
- Riemann integrable
- Continuous $\implies$ integrable
- The integral is linear
- The composition of a continuous function and integrable function is integrable.
- Fundamental theorems of calculus
- logarithm and exponentiation.
- change of variables

15. Interchanging limit operations.

- Uniform convergence and integrals
- Uniform convergence and differentiation.
- Leibniz rule(not examinable).

16. Series

- Definition
- Absolute convergence and conditional convergence
- Rearranging series
- Tests of convergence(Comparison, Root, Ratio, and **Integral**)

17. Power series

- Definition
- Examples from Taylor series
- Radius of convergence
- Weierstrass M-test
- Integral/Derivatives of powers series

Questions

Questions

1. Let X, Y , and Z be metric spaces and $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be maps. Assume further that:

- X is compact;
- f is surjective and continuous; and
- $g \circ f$ is continuous.

Show that g is continuous.

2. Determine whether the function

$$f(x) = |x|$$

is differentiable at $x = 0$.

3. Suppose f is differentiable on (a, b) and continuous on $[a, b]$. Prove that if

$$f'(x) > 1 \quad \text{for all } x \in (a, b),$$

then

$$f(b) - f(a) > b - a.$$

4. Let $f(x)$ be continuous at 0. Show that $xf(x)$ is differentiable at 0.
5. Prove the Integral Test for series convergence.
6. Prove that the following series converge:

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

(b) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

(c) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

(d) $\sum_{n=1}^{\infty} \frac{\sin(n)}{n^2}$

7. Compute the radius of convergence of the following power series:

(a) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

(b) $\sum_{n=1}^{\infty} \frac{3^n}{\sqrt{n}} x^n$

(c) $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} x^n$

(d) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{n^2} x^n$

8. Let f be a bounded integrable function from $[a, b] \rightarrow \mathbb{R}$. Prove that

$$F(x) := \int_a^x f(t) \, dt$$

is continuous on $[a, b]$.

9. Consider the infinite series of functions defined on the interval $[0, \pi]$ by:

$$S(x) = \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}$$

- (a) Prove that the series converges uniformly on $[0, \pi]$ to a continuous function $S(x)$. (*Hint: Use the Weierstrass M -test.*)
- (b) Justify why the integral and summation operations can be interchanged, and evaluate the integrated series explicitly:

$$\int_0^\pi \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2} \, dx = \sum_{n=1}^{\infty} \int_0^\pi \frac{\sin(nx)}{n^2} \, dx$$

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a uniformly continuous function, and let

$$f_n(x) = \frac{n}{2} \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} f(t) \, dt.$$

Show that $f_n \rightarrow f$ uniformly on $\mathbb{R}$.